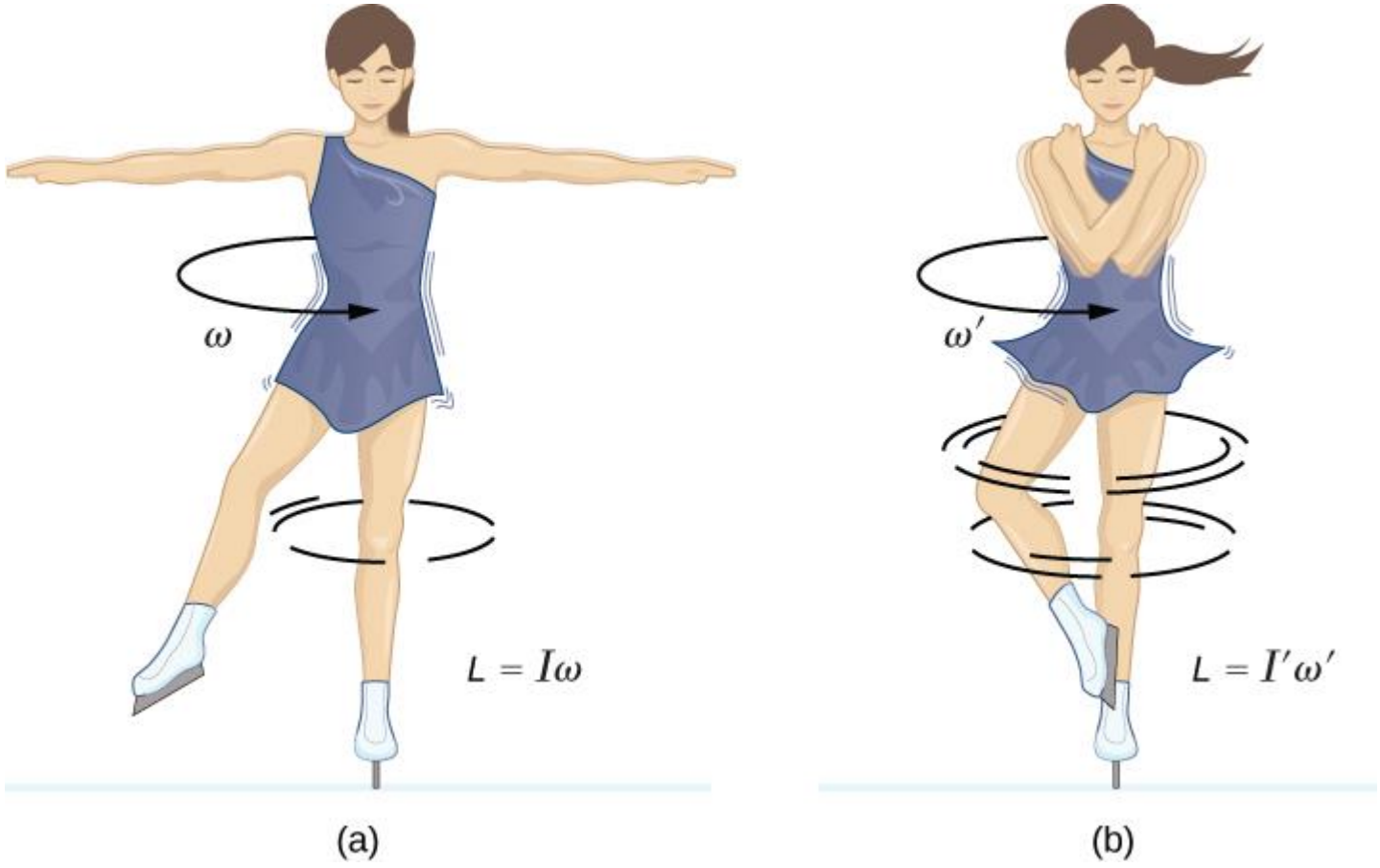
CHAPTER 11: MÔ MEN ĐỘNG LƯỢNG



https://pressbooks.online.ucf.edu/phy2048tjb/chapter/11-2-conservation-of-angular-momentum/

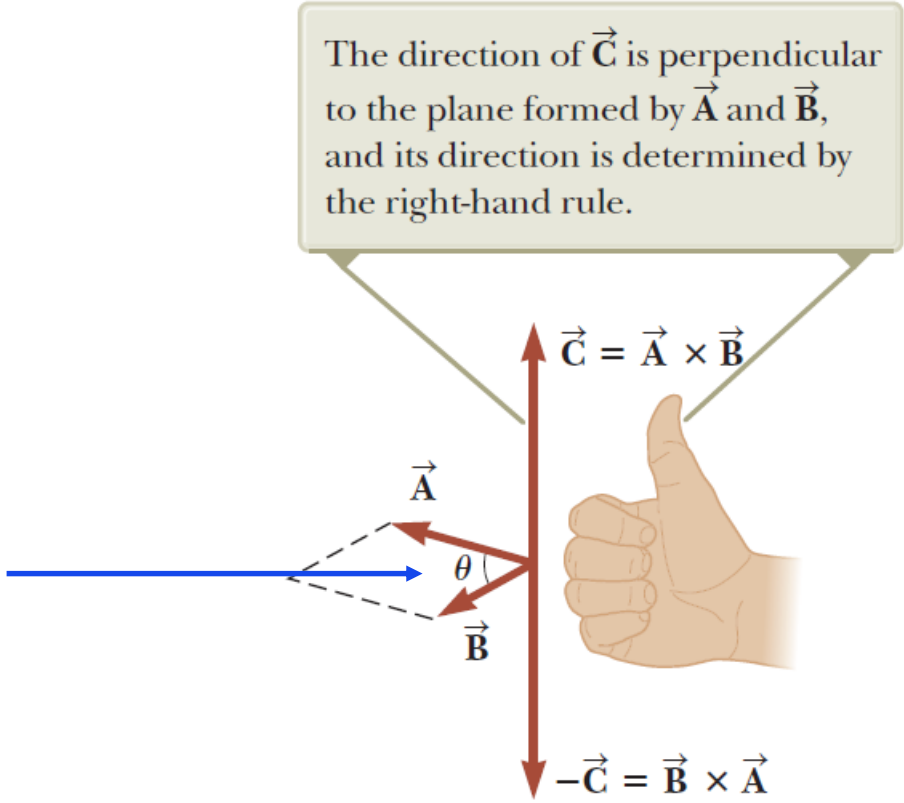
TÍCH CÓ HƯỚNG (cross-product)

- Tích có hướng giữa 2 vector $\vec{A}$ và $\vec{B}$

$$\vec{\mathbf{C}} = \vec{\mathbf{A}} \times \vec{\mathbf{B}}$$

Độ lớn $\boxed{C \equiv AB\sin\theta}$

θ : góc giữa $\vec{A}$ và $\vec{B}$



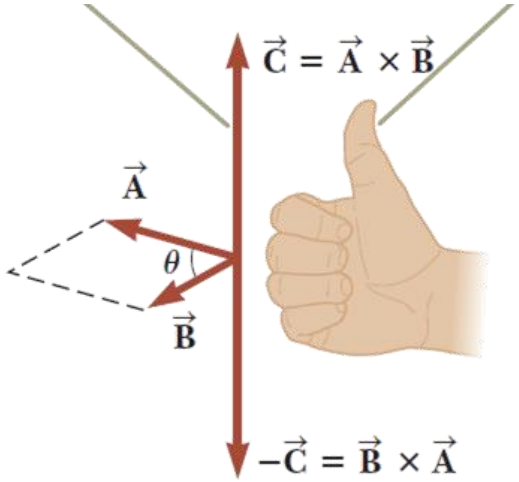
- Hướng vector $\vec{C}$ vuông góc với mặt phẳng tạo bởi $\vec{A}$ & $\vec{B}$
- Hướng của $\vec{C}$ được định nghĩa theo quy tắc bàn tay phải

CÁC TÍNH CHẤT CỦA TÍCH CÓ HƯỚNG

- Tích có hướng không có tính giao hoán

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

- Nếu $\vec{A}$ song song $\vec{B}$ (Θ = 0º or 180º), thì $\vec{A}$ x $\vec{B}$ = 0
  - $\vec{A}$ x $\vec{A}$= 0
- Nếu $\vec{A}$ vuông góc $\vec{B}$ ($\theta = 90^o$) thì |$\vec{A}$ x $\vec{B}$| = AB
- Tính chất phân phối:



$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

$$\frac{d}{dt}(\vec{A} \times \vec{B}) = \frac{d\vec{A}}{dt} \times \vec{B} + \vec{A} \times \frac{d\vec{B}}{dt}$$

Vector Mômen lực

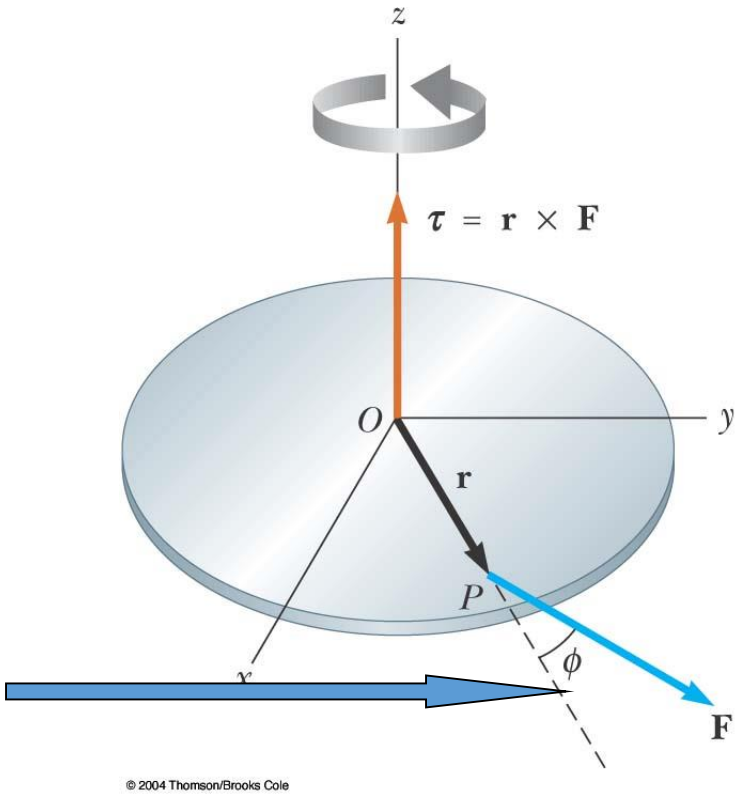
- Vector mômen lực $\vec{\tau}$

$$\vec{\boldsymbol{\tau}} \equiv \vec{\mathbf{r}} \times \vec{\mathbf{F}}$$

- Độ lớn:

$$\tau = rF \sin \phi$$

ϕ *góc giữa* $\vec{r}$ và $\vec{F}$



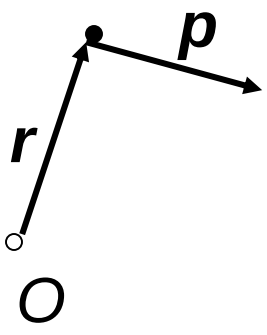
- Vector $\vec{\tau}$ nằm theo phương vuông góc với mặt phẳng tạo bởi vector vị trí $\vec{r}$ và lực tác dụng $\vec{F}$

MÔMEN ĐỘNG LƯỢNG

Chất điểm tại vị trí ***r*** (measured from point O) chuyển động với động lượng $\vec{p}=m\vec{v}$ có mômen động lượng:

$$\vec{\mathbf{L}} \equiv \vec{\mathbf{r}} \times \vec{\mathbf{p}}$$

Đơn vị: kg m^2/s



Độ lớn:

$$L = mvr \sin \phi$$

where ϕ is the angle between $\vec{\mathbf{r}}$ and $\vec{\mathbf{p}}$

Hệ gồm nhiều chất điểm có mômen động lượng toàn phần:

$$\vec{\mathbf{L}}_{\text{tot}} = \vec{\mathbf{L}}_1 + \vec{\mathbf{L}}_2 + \cdots + \vec{\mathbf{L}}_n = \sum_i \vec{\mathbf{L}}_i$$

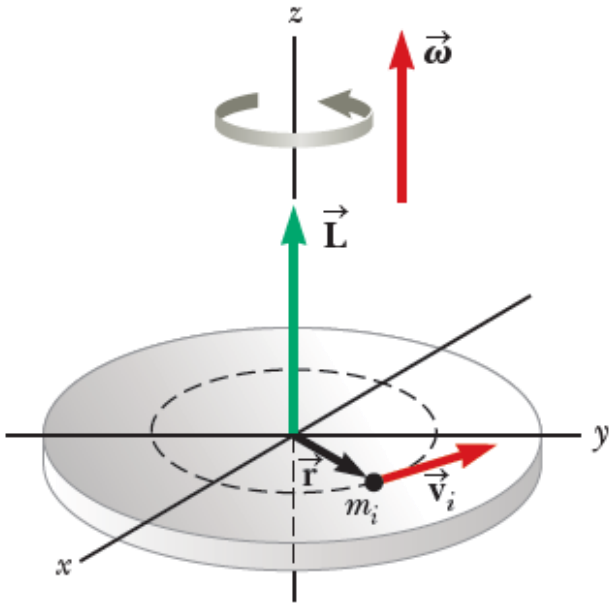
MÔMEN ĐỘNG LƯỢNG CỦA VẬT RẮN

Mômen động lượng:

$$\vec{L} = I\vec{\omega}$$

I: mômen quán tính

$\vec{\omega}$: vận tốc góc



The direction of $\vec{L}$ is the same as the direction of $\vec{\omega}$

HỆ KHÔNG CÔ LẬP VỀ MÔMEN ĐỘNG LƯỢNG

Tịnh tiến

Hệ không cô lập về động lượng:

Định luật II Newton:

$$\sum \vec{\mathbf{F}} = d\vec{\mathbf{p}}/dt$$

Quay

$$\vec{\boldsymbol{\tau}} \equiv \vec{\mathbf{r}} \times \vec{\mathbf{F}}$$

+ Hệ một hạt (chất điểm)

$$\sum \vec{\boldsymbol{\tau}} = \frac{d\vec{\mathbf{L}}}{dt}$$

$$\vec{\mathbf{L}} \equiv \vec{\mathbf{r}} \times \vec{\mathbf{p}}$$

+ Hệ nhiều chất điểm

$$\sum \vec{\boldsymbol{\tau}}_{\text{ext}} = \frac{d\vec{\mathbf{L}}_{\text{tot}}}{dt}$$

+ Đối với vật rắn:

$$\sum \tau_{\text{ext}} = I\alpha$$

$$\vec{L} = I\vec{\omega}$$

HỆ CÔ LẬP VỀ MÔMEN ĐỘNG LƯỢNG

Hệ cô lập về ĐỘNG LƯỢNG

- Nếu tổng ngoại lực tác dụng lên hệ bằng 0

$\Rightarrow$ Động lượng được bảo toàn:

$$\vec{p}_i = \vec{p}_f$$

Hệ cô lập về MÔMEN ĐỘNG LƯỢNG

- Nếu tổng mômen lực tác dụng lên hệ bằng 0

=> Mô men động lượng bảo toàn

$$\vec{L}_i = \vec{L}_f$$

SO SÁNH CĐ TỊNH TIẾN VÀ CĐ QUAY

Đại lượng	**Tịnh tiến**	**Quay**
Vị trí	$\boldsymbol{x}$	θ
Tốc độ	$\boldsymbol{v}$	ω
Gia tốc	$\boldsymbol{a}$	α
Quán tính	m	$I \sim (\text{constant})mr^2$
Động năng	$K_{trans} = 1/2\ mv^2$	$K_{rot} = 1/2\ I\omega^2$
Động lượng	$\boldsymbol{p} = m\boldsymbol{v}$	$\boldsymbol{L} = I\omega$
ĐL II Newton	$\Sigma\ \boldsymbol{F} = d\boldsymbol{p}/dt$	$\Sigma\ \tau = dL/dt$
Công	$W = F_{\parallel}\ \Delta x$	$W = \tau\ \Delta\theta$
Bảo toàn	$\Delta\boldsymbol{p} = 0$ if $\Sigma\boldsymbol{F}_{ext}=0$	$\Delta\boldsymbol{L} = 0$ if $\Sigma\tau_{ext}=0$
Xung	$\boldsymbol{F}\Delta t = \Delta\boldsymbol{p}$	$\tau\ \Delta t = \Delta\boldsymbol{L}$

ĐỊNH NGHĨA HỆ CÔ LẬP

	Năng lượng	**Động lượng**	**Mômen động lượng**
Định nghĩa	Không có sự trao đổi năng lượng qua biên giới hệ	Tổng ngoại lực tác dụng lên hệ bằng 0	Tổng mômen lực tác dụng lên hệ bằng 0
Bảo toàn	E	p	L
Phương trình	$E_i = E_f$	$\vec{p}_i = \vec{p}_f$	$\vec{L}_i = \vec{L}_f$

Example 11.4 A System of Objects AM

A sphere of mass m_1 and a block of mass m_2 are connected by a light cord that passes over a pulley as shown in Figure 11.6. The radius of the pulley is R, and the mass of the thin rim is M. The spokes of the pulley have negligible mass. The block slides on a frictionless, horizontal surface. Find an expression for the linear acceleration of the two objects, using the concepts of angular momentum and torque.

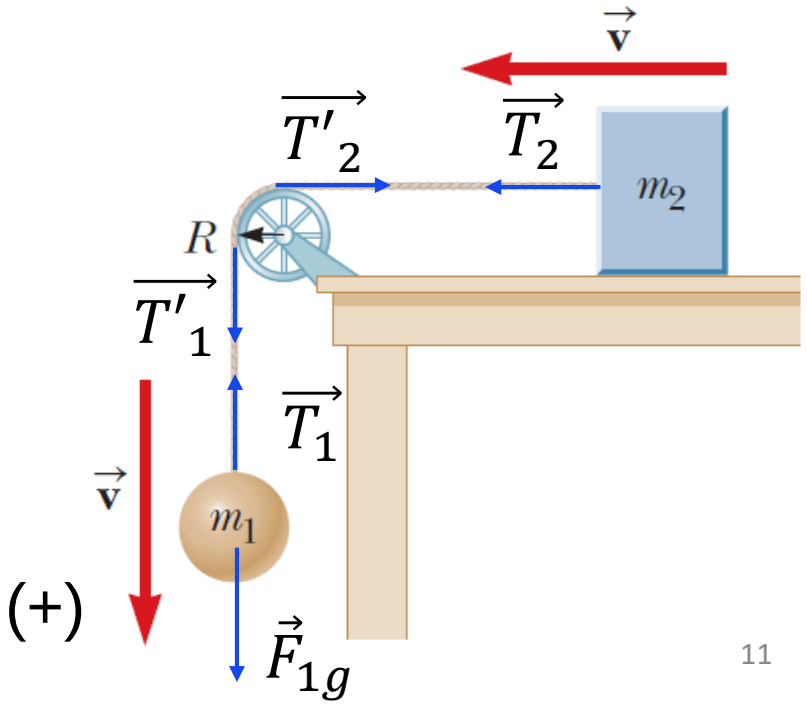
Solutions

Way 1: Apply Newton's second law as in Chapter 5

Apply Newton's second law for m1:

$$\sum \vec{F}_1 = m_1\vec{a}_1 \Leftrightarrow \vec{T}_1 + \vec{F}_{1g} = m_1\vec{a}_1$$

(+) $\downarrow$ $m_1 g - T_1 = m_1 a_1$ (1)



Apply Newton's second law for m2:

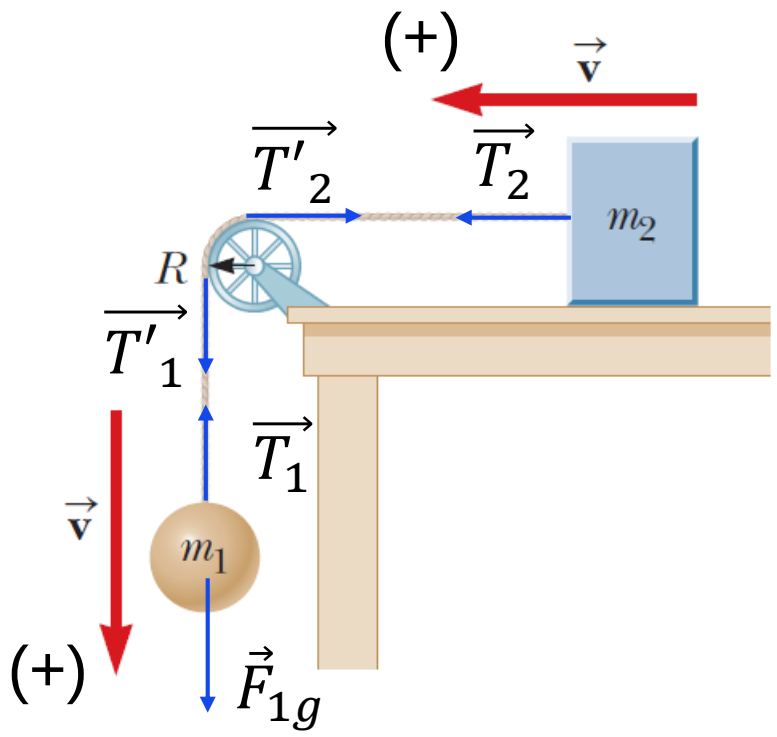
$$\sum \vec{F}_2 = m_2\vec{a}_2 \Leftrightarrow \vec{T}_2 = m_2\vec{a}_2$$

(+) $\leftarrow$ $T_2 = m_2 a_2 \quad (2)$

Apply Newton's second law for the pulley:

$$\sum \vec{F}_{pulley} = M\vec{a}_t \Leftrightarrow \vec{T}'_1 + \vec{T}'_2 = M\vec{a}_t$$

<=> $T'_1 - T'_2 = Ma_t$ (3)



The mass of the cord is negligible => $T_1 = T'_1$, $T_2 = T'_2$

The cord is unstretched => $a_1 = a_2 = a_t$

Eq (1),(2) and(3) becomes:

$$m_1 g - T_1 = m_1 a \ (1) \quad \Rightarrow \quad T_1 = m_1 g - m_1 a \ (1')$$

$$T_2 = m_2 a \ \ (2)$$

$$T_1 - T_2 = Ma \ (3)$$

Substituting eq(1') and(2) into eq(3):

$$m_1 g - m_1 a \ - m_2 a = Ma \ (3')$$

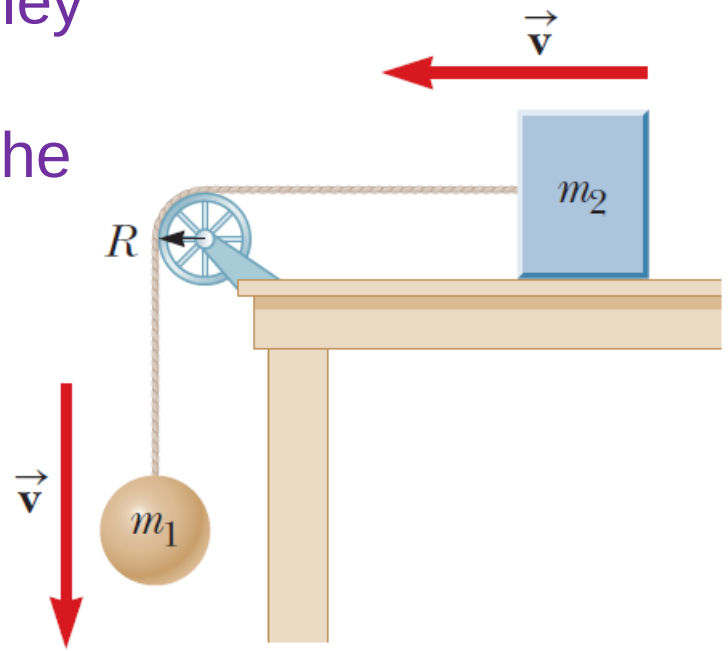
$$\Leftrightarrow m_1 g = (m_1 + m_2 + M)a$$

$$\Leftrightarrow a = \frac{m_1 g}{m_1 + m_2 + M} \ \ (\mathrm{m/s^2})$$

Way 2: Apply the linear momentum approach

Consider the system of sphere, block, cord, and pulley

⇒ This is an nonisolated system in terms of linear momentum since the net external force apply to the system is different from zero ($F_1g \neq 0$):



$$\sum \vec{\mathbf{F}} = d\vec{\mathbf{p}}/dt$$

$$\vec{F}_{1g} = \frac{d\vec{p}}{dt}$$

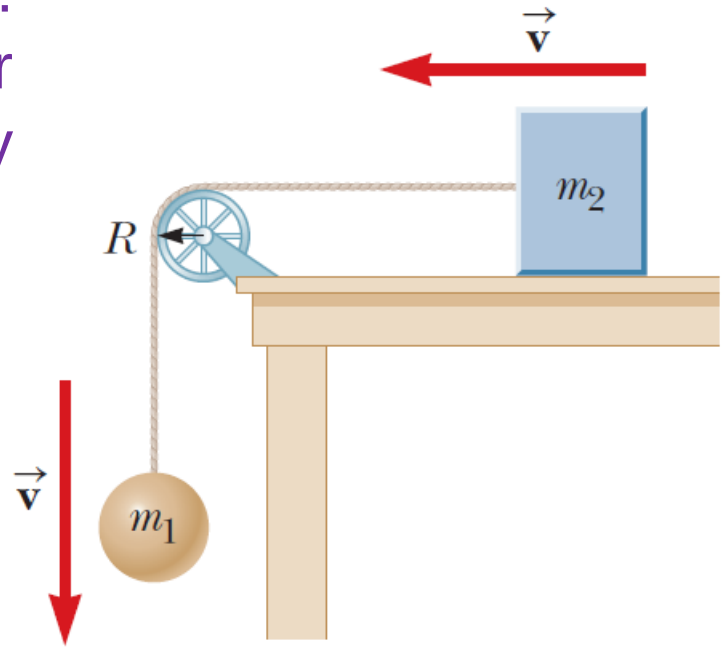
$$\Leftrightarrow m_1 g = \frac{d}{dt}(m_1 v + m_2 v + M v)$$

$$\Leftrightarrow m_1 g = (m_1 + m_2 + M)\frac{dv}{dt} = (m_1 + m_2 + M)a$$

$$\Leftrightarrow a = \frac{m_1 g}{m_1 + m_2 + M} \text{ (m/s}^2\text{)}$$

Way 3: Apply the concepts of angular momentum and torque

Consider the system of pulley, m1, m2 and cord: this is non-isolated system in terms of angular momentum due to the external torque caused by gravitational force F_{1g}



$$\sum \vec{\tau}_{\text{ext}} = \frac{d\vec{\mathbf{L}}_{\text{tot}}}{dt}$$

$$\vec{F}_{g1} \text{x} \vec{R} = \frac{d\vec{L}}{dt} = \frac{d(\vec{R} \text{ x } \vec{p})}{dt}$$

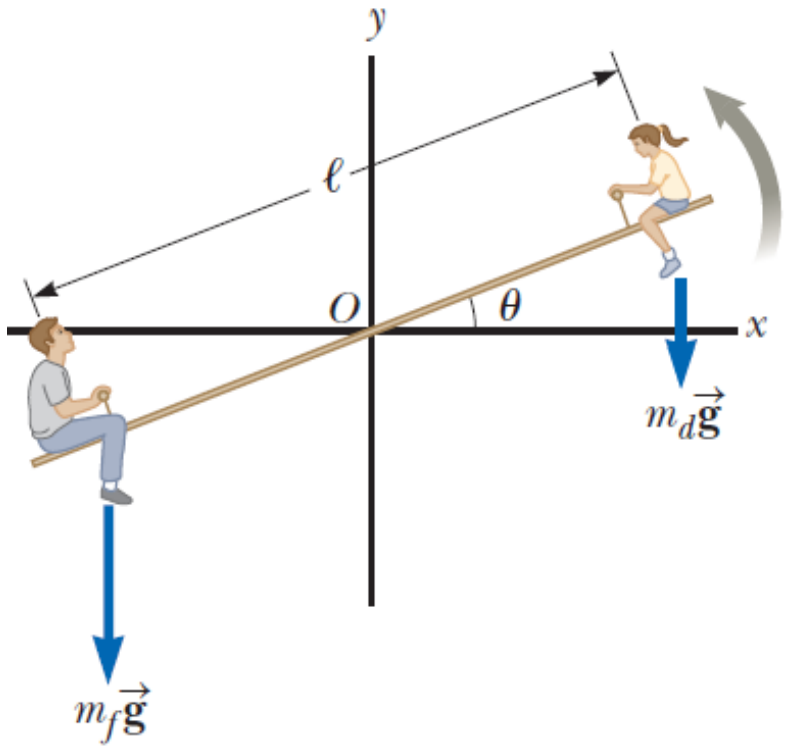
$$\Leftrightarrow m_1 g R = \text{R} \frac{d(m_1 v + m_2 v + M v)}{dt} = R(m_1 + m_2 + M)\frac{dv}{dt}$$

$$\Leftrightarrow m_1 g = (m_1 + m_2 + M)a \Rightarrow a = \frac{m_1 g}{m_1 + m_2 + M} \quad (\text{m/s}^2)$$

Example 11.6 The Seesaw AM

A father of mass m_f and his daughter of mass m_d sit on opposite ends of a seesaw at equal distances from the pivot at the center (Fig. 11.9). The seesaw is modeled as a rigid rod of mass M and length ℓ and is pivoted without friction. At a given moment, the combination rotates in a vertical plane with an angular speed ω.

(A) Find an expression for the magnitude of the system's angular momentum.



Solutions

Angular momentum: $L = I\omega$ (1)

Moment of inertia: I = $\sum m_i r_i^2 = m_f\left(\frac{l}{2}\right)^2 + m_d\left(\frac{l}{2}\right)^2 + (\sum m_k r_k^2)_{seesaw}$

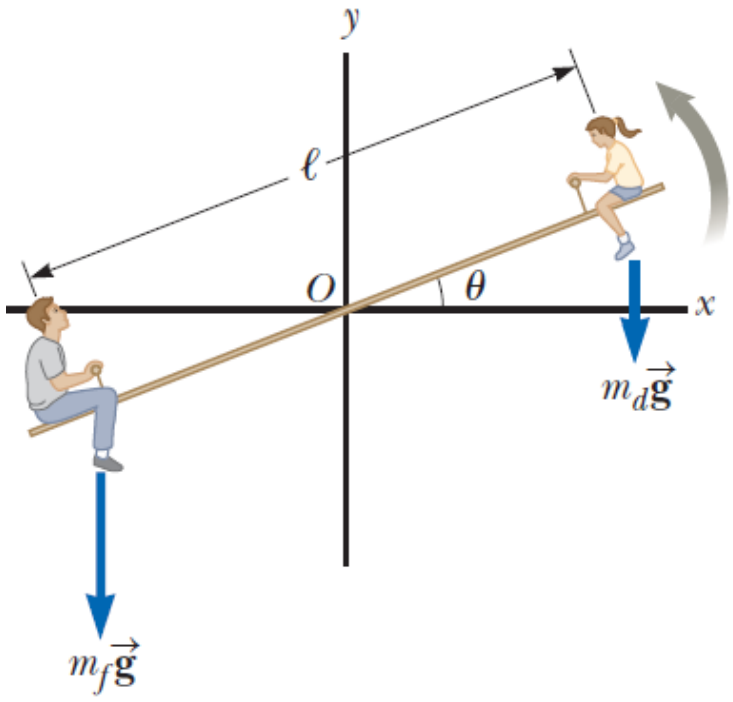
=> $I = m_f\left(\frac{l}{2}\right)^2 + m_d\left(\frac{l}{2}\right)^2 + \frac{1}{12}Ml^2$ (2)

Substituting eq(2) into eq(1)

$$L = I\omega$$

$$L = \left[m_f\left(\frac{l}{2}\right)^2 + m_d\left(\frac{l}{2}\right)^2 + \frac{1}{12}Ml^2 \right]\omega$$

$$= \frac{l^2}{4}\left(m_f + m_d + \frac{M}{3}\right)\omega$$



(B) Find an expression for the magnitude of the angular acceleration of the system when the seesaw makes an angle θ with the horizontal.

Solutions

$$\sum \tau_{\text{ext}} = I\alpha$$

$\Leftrightarrow \tau_f + \tau_d = I\alpha$

F_{fg} cause the rotation counterclockwise

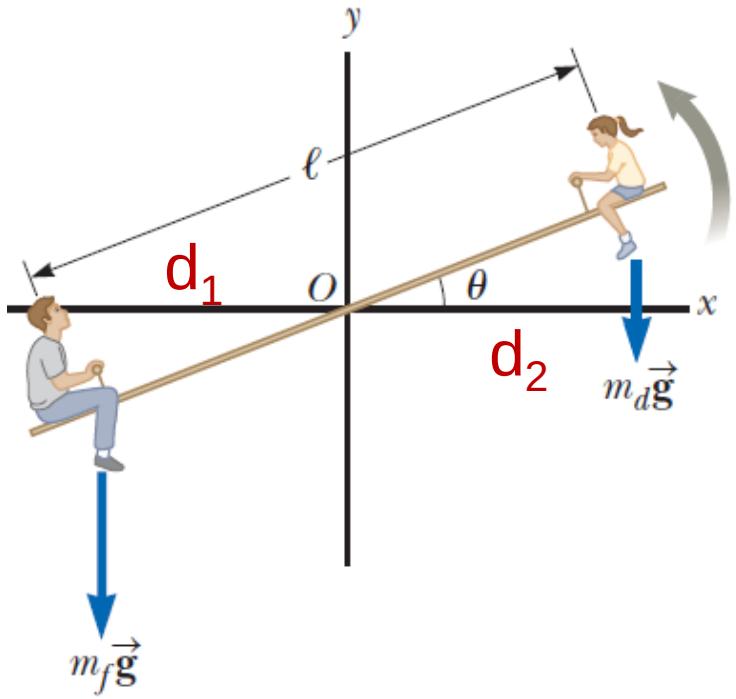
$\Rightarrow \tau_f > 0$

F_{dg} cause the rotation clockwise => $\tau_d < 0$

$\Rightarrow \tau_f - \tau_d = I\alpha$

$\Leftrightarrow m_f g d_1 - m_d g d_2 = I\alpha$

$\Leftrightarrow m_f g \frac{l}{2} cos\theta - m_d g \frac{l}{2} cos\theta = I\alpha$

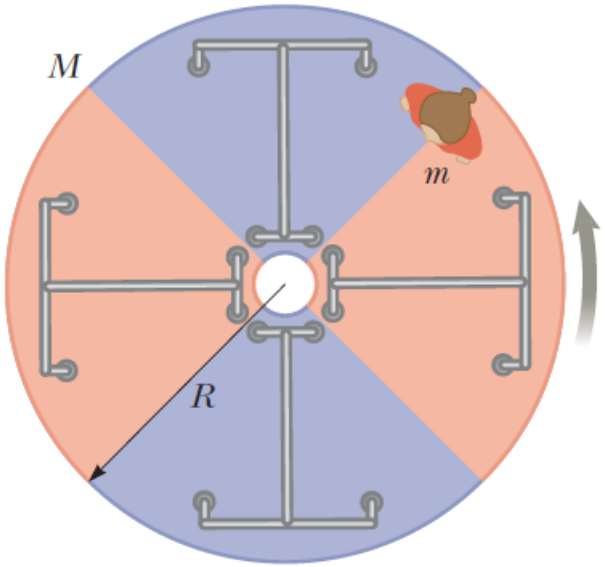


$$\alpha = \frac{(m_f - m_d)}{\frac{l^2}{4}\left(m_f + m_d + \frac{M}{3}\right)} g \frac{l}{2} cos\theta$$

$$= \frac{2g(m_f - m_d)cos\theta}{l\left(m_f + m_d + \frac{M}{3}\right)}$$

Example 11.8 The Merry-Go-Round AM

A horizontal platform in the shape of a circular disk rotates freely in a horizontal plane about a frictionless, vertical axle (Fig. 11.11). The platform has a mass $M = 100$ kg and a radius $R = 2.0$ m. A student whose mass is $m = 60$ kg walks slowly from the rim of the disk toward its center. If the angular speed of the system is 2.0 rad/s when the student is at the rim, what is the angular speed when she reaches a point $r = 0.50$ m from the center?



Solutions

Consider the system of the student and the platform is isolated system in terms of angular momentum

$\Rightarrow$ The angular momentum is conserved:

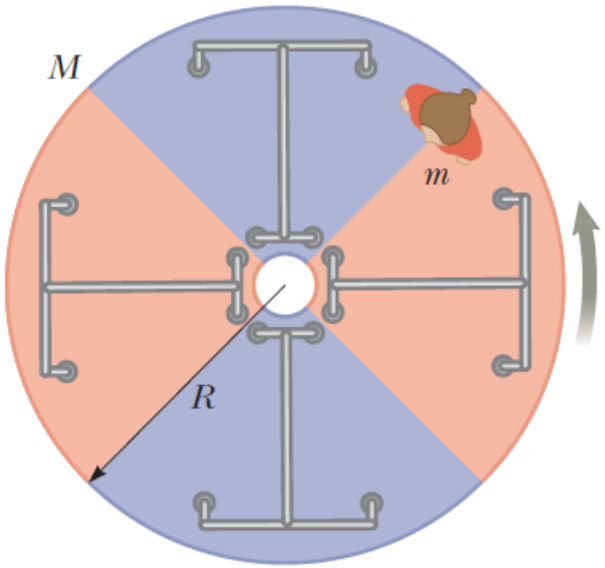
$$\Rightarrow \vec{L}_i = \vec{L}_f$$

$$\Leftrightarrow I_i\omega_i = I_f\omega_f$$

$$\Leftrightarrow (I_{disk} + I_{student_i})\omega_i = (I_{disk} + I_{student_f})\omega_f$$

$$=> \left(\frac{1}{2}MR^2 + mR^2\right)\omega_i = \left(\frac{1}{2}MR^2 + m(r)^2\right)\omega_f$$

$$=> \omega_f = \frac{\left(\frac{1}{2}MR^2 + mR^2\right)\omega_i}{\frac{1}{2}MR^2 + mr^2} = 4.1 \text{ rad/s}$$



Solutions

The system of two disks is isolated system in terms of angular momentum

$\Rightarrow$ Conservation of angular momentum:

$$\Rightarrow \vec{L}_i = \vec{L}_f$$

$$\Leftrightarrow I_i\omega_i = I_f\omega_f$$

$$\Leftrightarrow I_1\omega_i = (I_1 + I_2)\omega_f$$

$$\Leftrightarrow \omega_f = \frac{I_1}{I_1+I_2}\omega_i$$

(b) The ratio between the final and initial rotational energy:

$$\frac{K_f}{K_i} = \frac{1/2 I_f {\omega_f}^2}{1/2 I_i {\omega_i}^2} = \frac{I_1 + I_2}{I_1 {\omega_i}^2}\left(\frac{I_1}{I_1 + I_2}\omega_i\right)^2 = \frac{I_1}{I_1 + I_2}$$

30. A disk with moment of inertia I_1 rotates about a frictionless, vertical axle with angular speed ω_i. A second disk, this one having moment of inertia I_2 and initially not rotating, drops onto the first disk (Fig. P11.30). Because of friction between the surfaces, the two eventually reach the same angular speed ω_f. (a) Calculate ω_f. (b) Calculate the ratio of the final to the initial rotational energy.

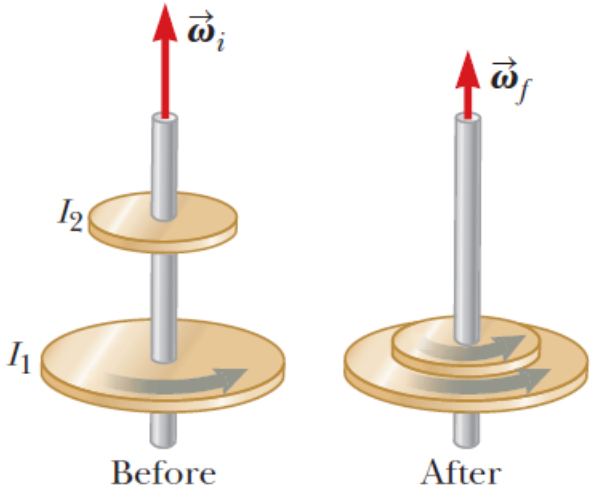


Figure P11.30

HOMEWORK

11. M A light, rigid rod of length $\ell = 1.00$ m joins two particles, with masses $m_1 = 4.00$ kg and $m_2 = 3.00$ kg, at its ends. The combination rotates in the xy plane about a pivot through the center of the rod (Fig. P11.11). Determine the angular momentum of the system about the origin when the speed of each particle is 5.00 m/s.

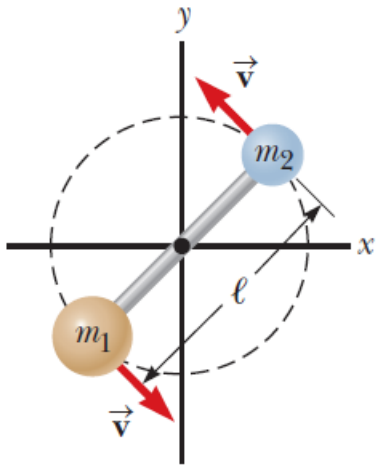


Figure P11.11

18. AMT W A counterweight of mass $m = 4.00$ kg is attached to a light cord that is wound around a pulley as in Figure P11.18. The pulley is a thin hoop of radius $R = 8.00$ cm and mass $M = 2.00$ kg. The spokes have negligible mass. (a) What is the magnitude of the net torque on the system about the axle of the pulley? (b) When the counterweight has a speed v, the pulley has an angular speed $\omega = v/R$. Determine the magnitude of the total angular momentum of the system about the axle of the pulley. (c) Using your result from part (b) and $\vec{\tau} = d\vec{\mathbf{L}}/dt$, calculate the acceleration of the counterweight.

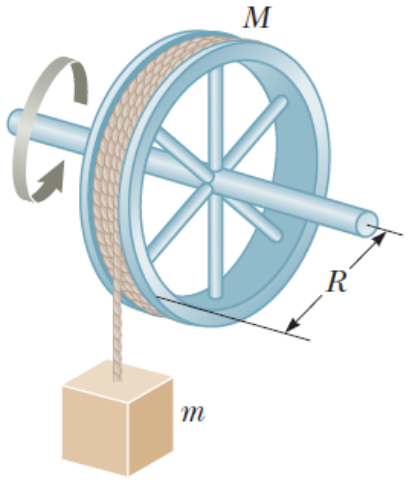


Figure P11.18

HOMEWORK

34. W A student sits on a freely rotating stool holding two dumbbells, each of mass 3.00 kg (Fig. P11.34). When his arms are extended horizontally (Fig. P11.34a), the dumbbells are 1.00 m from the axis of rotation and the student rotates with an angular speed of 0.750 rad/s. The moment of inertia of the student plus stool is 3.00 kg · m^2 and is assumed to be constant. The student pulls the dumbbells inward horizontally to a position 0.300 m from the rotation axis (Fig. P11.34b). (a) Find the new angular speed of the student. (b) Find the kinetic energy of the rotating system before and after he pulls the dumbbells inward.

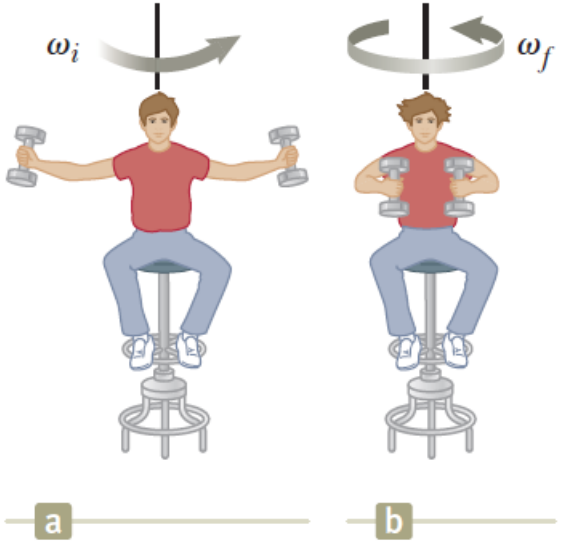


Figure P11.34

52. AMT M A puck of mass $m = 50.0$ g is attached to a taut cord passing through a small hole in a frictionless, horizontal surface (Fig. P11.52). The puck is initially orbiting with speed $v_i = 1.50$ m/s in a circle of radius $r_i = 0.300$ m. The cord is then slowly pulled from below, decreasing the radius of the circle to $r = 0.100$ m. (a) What is the puck's speed at the smaller radius? (b) Find the tension in the cord at the smaller radius. (c) How much work is done by the hand in pulling the cord so that the radius of the puck's motion changes from 0.300 m to 0.100 m?

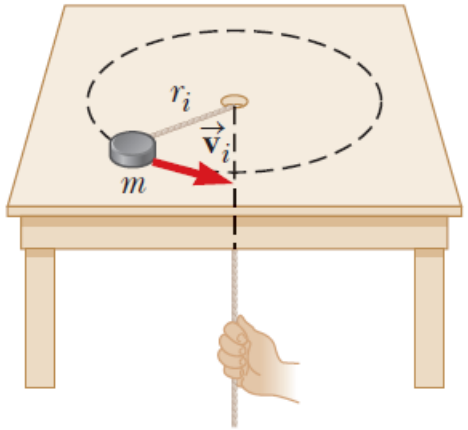


Figure P11.52 Problems 52 and 53.

HOMEWORK

60. The puck in Figure P11.60 has a mass of 0.120 kg. The distance of the puck from the center of rotation is originally 40.0 cm, and the puck is sliding with a speed of 80.0 cm/s. The string is pulled downward 15.0 cm through the hole in the frictionless table. Determine the work done on the puck. (*Suggestion:* Consider the change of kinetic energy.)

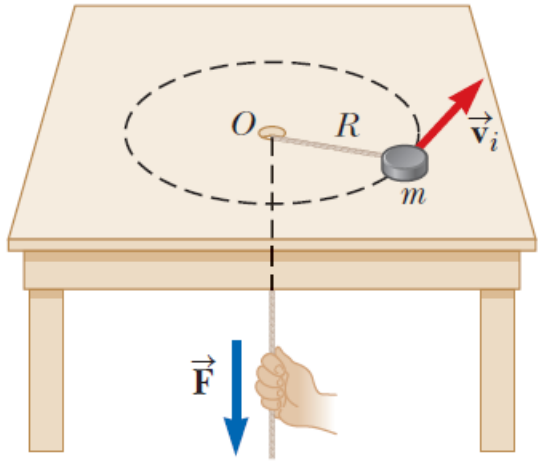


Figure P11.60